BASSAM WALID ALJAZZAR

Lead AI & Mechatronics Systems Engineer | US Patent Holder |
Security Technology & Public Safety Innovation
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PROFILE

AI & Mechatronics Systems Engineer with 8+ years leading end-to-end development of security, public safety, and automation systems — from concept brief to field-tested pilot. US Patent holder (US11573635) with hands-on expertise across computer vision, edge AI, embedded systems, and hardware fabrication. At the UAE Ministry of Interior Innovation Lab, led 8 officially published innovation projects, delivered two systems to active field pilots with UAE police units, represented MOI at 5 international security exhibitions (ISNR 2022 & 2024; World Police Summit 2023–2025), and is currently developing an AI wearable security proof of concept for ISNR Abu Dhabi 2026.

Targeting: Senior / Lead Engineer roles in AI systems, robotics, or industrial automation.

CORE STRENGTHS

AI Systems Architecture
Computer Vision · Edge AI · LLM Integration · Voice AI

Embedded & Robotics
ESP32 · Raspberry Pi · ROS · Sensor Fusion · MQTT

Prototype to Field Pilot
3D Design · FDM/SLS · Python/Web Stack · POC → Deployment

WORK EXPERIENCE

Lead Mechatronics & AI Engineer

Jun 2021 – Present

Tatweer Middle East & Africa — UAE Ministry of Interior, Manufacturing Innovation Lab | Abu Dhabi, UAE

ROLE & OWNERSHIP

- Primary technical lead for AI and mechatronic innovation initiatives at the MOI Manufacturing Innovation Lab - owning system architecture, embedded integration, software stack, prototype delivery, and stakeholder presentation across projects.
- Managed full project lifecycle end-to-end: problem framing, system design, prototype build, internal review, and official publication on the MOI Think10X platform.

IMPACT & DELIVERY

- 8 innovation projects officially published by UAE Ministry of Interior — spanning AI vision, robotics, traffic intelligence, and security automation (think10x.moi.gov.ae).
- 2 systems reached field pilot stage: Leo Rover Inspection Platform delivered to UAE Explosives Police for vehicle inspection trials; BiSOS Smart Patrol Device delivered to UAE Community Police for field testing.
- Exhibition Record — Represented UAE Ministry of Interior Innovation Lab as exhibitor at 5 major international security events: ISNR Abu Dhabi 2022 & 2024, and World Police Summit 2023, 2024 & 2025 — presenting government innovation projects to international security and law enforcement audiences.
- Delivered 3–4 technical workshops annually since 2021 (15+ workshops total) covering 3D design, electronics, Arduino, and additive manufacturing — including instruction in the MOI 'Mubtakir' Innovation Diploma program.

FLAGSHIP SYSTEMS

- **Smart Security Vest**: Wearable AI platform combining dual-camera weapon detection, face recognition, CO sensing, mmWave biometrics, and a live operations dashboard. Currently being developed for presentation at ISNR Abu Dhabi 2026.
- **AI Police Reporting Assistant**: Arabic voice-interactive reporting system using OpenAI Realtime API to generate structured accident reports from voice input and scene images; in active internal use at MOI Innovation Lab for accident reporting and documentation workflows.
- **Electronic Shooting Trainer:** Laser-based marksmanship system with custom-invented LDR sensor array (10–15cm laser collection radius); automated performance evaluation and qualification tracking.
- **Autonomous Perimeter Security System**: AI-triggered breach detection system that launches a drone, tracks the intruder, and relays image plus GPS coordinates to the operations room in real time.
- **Smart Inspection Platform**: Mobile multi-camera proof of concept for rapid hazardous-object detection at temporary field events and checkpoints.
- **BiSOS Smart Patrol Device**: ESP32-based collision detection system for police scooter officers with 10-second automatic alerting to ambulance operations, GPS location sharing, and a 4-level MQTT priority queue.

Research Engineer

May 2019 – May 2021

UAE University, College of Information Technology | Al Ain, UAE

- Led full development of Bu Saif Humanoid Robot — Arabic NLP, silicone facial actuators, servo-driven arms; showcased at GITEX 2019, Dubai.
- Developed MMSE cognitive assessment program on Bu Saif (automated patient scoring + response-time analysis); outcome: institutional MOU signed with Dubai Health Authority for elderly cognitive care development.
- Invented Meta Touch contactless elevator interface during COVID-19 — product commercially deployed in UAE market.
- Co-invented US Patent US11573635 (Feb 2023): EMG-based communication face mask for patients with motor disabilities. Assigned to UAE University.
- Produced low-cost 3D-printed ventilator prototypes during COVID-19 equipment shortage; logistics lead for JSSR2020 International Social Robotics Symposium.

Mechatronics Engineer

Nov 2018 – Apr 2019

Fadel Al Dhaheri General Trading | Dubai, UAE

- Maintenance and repair of embedded systems, automated machinery, and electronic control boards.

Mechatronics Engineer

Jun 2017 – Nov 2017

Flash Tech Company | Gaza, Palestine

- PLC programming, industrial maintenance design, solar energy system installation, and electrical load calculations.

TECHNICAL SKILLS

AI & Computer Vision	YOLO v5/v8 · OpenCV · Edge AI · OpenAI Realtime API · LLM Integration · Voice AI
Embedded Systems	Python (Expert) · Arduino/C++ · ESP32 · Raspberry Pi · XIAO ESP32-S3/C3 · MQTT · I^2C/SPI/UART · GPS/GSM
Robotics & Control	ROS / ROS2 · PyQt5 · Sensor Fusion · Motor Control · Real-time Systems
Design & Fabrication	SolidWorks (Expert) · Fusion 360 · AutoCAD · FDM/SLS 3D Printing · Laser Cutting · CNC · PCB Design

Software & Platforms	Firebase · n8n Automation · Linux/Ubuntu · Git · Docker · Ubuntu Touch

EDUCATION

BSc – Mechatronics Engineering **2011 – 2017**
Al-Azhar University | Gaza, Palestine

Graduation Project: Shrimp Rover Robot — 6-wheel independent-motor rocker-bogie mechanism for obstacle traversal.

PATENTS, RECOGNITION & KEY OUTCOMES

- **US Patent (2023)** Co-inventor of US Patent No. 11,573,635 for an EMG-based communication face mask; assigned to UAE University.
- **Government-Published Innovations** Eight innovation projects officially reviewed and published by the UAE Ministry of Interior on the national Think10X innovation platform.
- **Field Pilots** Two systems reached active pilot stage: Leo Rover with UAE Explosives Police and BiSOS with UAE Community Police.
- **GITEX 2019** Bu Saif Humanoid Robot presented at GITEX Technology Week, Dubai.
- **Institutional MOU** signed between UAE University and Dubai Health Authority for Bu Saif elderly cognitive-care deployment — outcome of research led at UAE University Innovation Lab.
- **Commercial Product** Meta Touch contactless elevator interface progressed to commercial deployment in the UAE market.
- **Conference Leadership** Organizer and logistics lead for JSSR2020, the 5th Joint UAE Symposium on Social Robotics.
- **Technical Instruction** — Instructor for 3D Design and Additive Manufacturing in the Mubtakir Innovation Diploma, delivering 3–4 workshops annually since 2021 (15+ total).

LANGUAGES

Arabic — Native Speaker

English — Advanced (Professional Working)

Portfolio & innovation references: linkedin.com/in/bassam-aljazzar · github.com/bassamaljazzar93 · think10x.moi.gov.ae